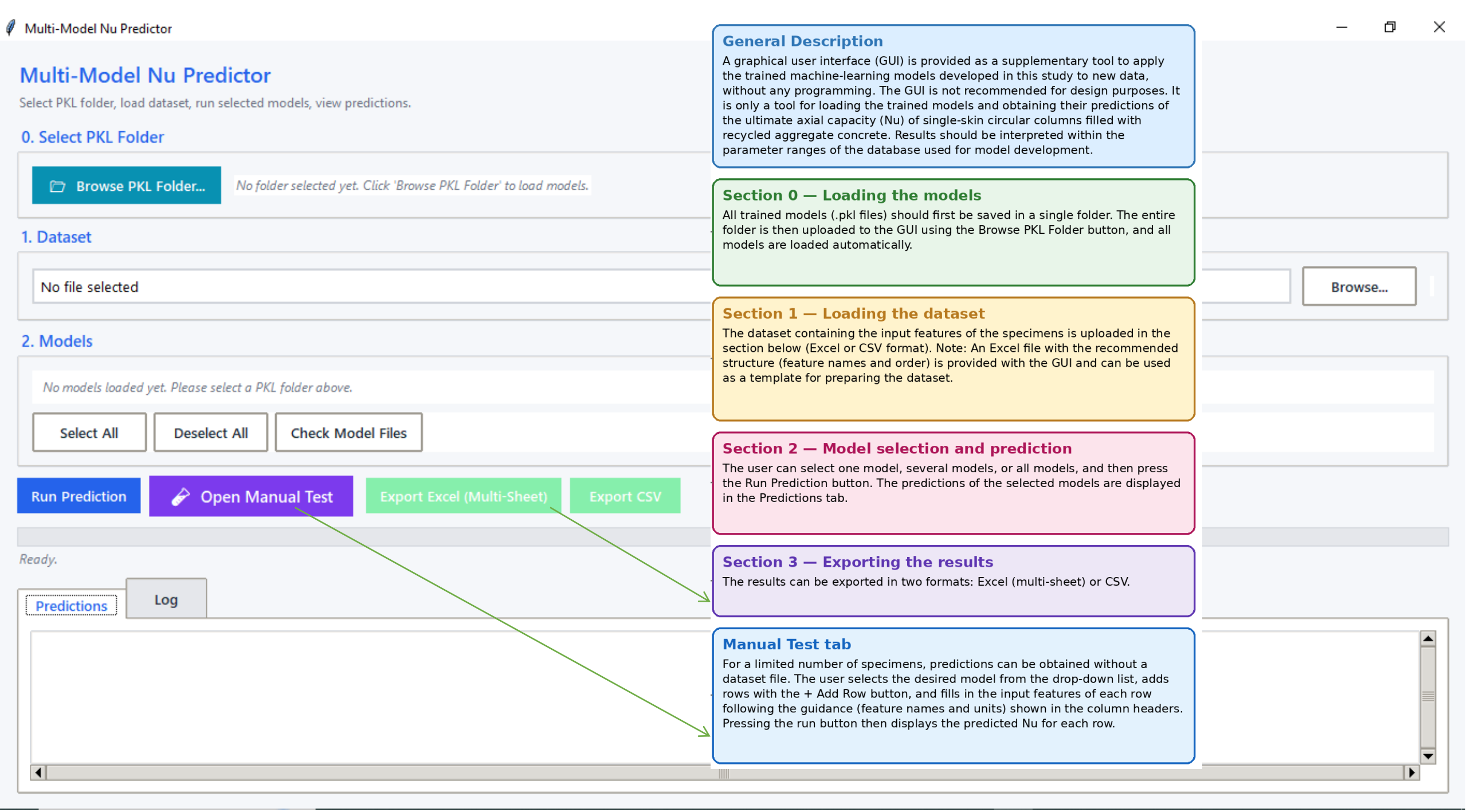

Multi-Model Nu Predictor
Multi-Model Nu Predictor
Select PKL folder, load dataset, run selected models, view predictions.
0. Select PKL Folder
Browse PKL Folder...
No folder selected yet. Click 'Browse PKL Folder' to load models.
1. Dataset
No file selected
Browse...
2. Models
No models loaded yet. Please select a PKL folder above.
Select All
Deselect All
Check Model Files
Run Prediction
Open Manual Test
Export Excel (Multi-Sheet)
Export CSV
Ready.
Predictions
Log
General Description
A graphical user interface (GUI) is provided as a supplementary tool to apply the trained machine-learning models developed in this study to new data, without any programming. The GUI is not recommended for design purposes. It is only a tool for loading the trained models and obtaining their predictions of the ultimate axial capacity (Nu) of single-skin circular columns filled with recycled aggregate concrete. Results should be interpreted within the parameter ranges of the database used for model development.
Section 0 — Loading the models
All trained models (.pkl files) should first be saved in a single folder. The entire folder is then uploaded to the GUI using the Browse PKL Folder button, and all models are loaded automatically.
Section 1 — Loading the dataset
The dataset containing the input features of the specimens is uploaded in the section below (Excel or CSV format). Note: An Excel file with the recommended structure (feature names and order) is provided with the GUI and can be used as a template for preparing the dataset.
Section 2 — Model selection and prediction
The user can select one model, several models, or all models, and then press the Run Prediction button. The predictions of the selected models are displayed in the Predictions tab.
Section 3 — Exporting the results
The results can be exported in two formats: Excel (multi-sheet) or CSV.
Manual Test tab
For a limited number of specimens, predictions can be obtained without a dataset file. The user selects the desired model from the drop-down list, adds rows with the + Add Row button, and fills in the input features of each row following the guidance (feature names and units) shown in the column headers. Pressing the run button then displays the predicted Nu for each row.